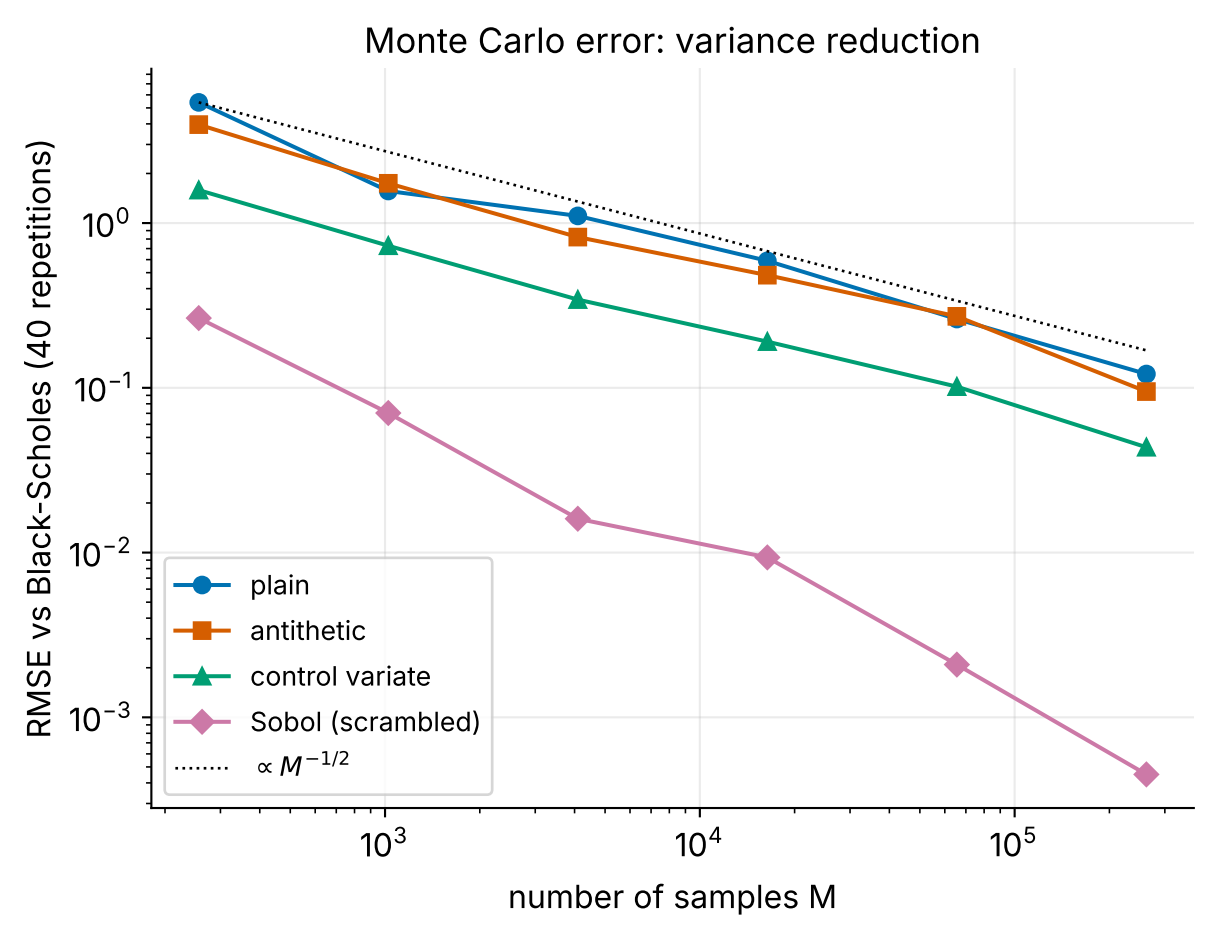
Monte Carlo error: variance reduction
RMSE vs Black-Scholes (40 repetitions)
number of samples M
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{3}
10^{4}
10^{5}
plain
antithetic
control variate
Sobol (scrambled)
$\propto M^{-1/2}$